Constant Acceleration: Vectors

Questions 1–31

Question 1. The points A and B have position vectors of $\begin{pmatrix}-2\\4\end{pmatrix}$ metres and $\begin{pmatrix}8\\11\end{pmatrix}$ metres respectively. Find the distance between A and B.

Question 2. Given that a particle travels at the constant velocity given, find the speed of the particle:

a) $(-2\mathbf{i}+4\mathbf{j})$ ms^{-1}

b) $\begin{pmatrix}5\\8\end{pmatrix}$ ms^{-1}

c) $(2.5k\mathbf{i}-6k\mathbf{j})$ ms^{-1}, where $k>0$

Question 3. A particle travels with a constant acceleration of $(-2\mathbf{i} + 3\mathbf{j})$ ms^{-2} from point A to point B. Its total displacement is $(-400\mathbf{i} + 250\mathbf{j})$ m. Its final velocity is $(12\mathbf{i} + 10\mathbf{j})$ ms^{-1}. Find

a) the magnitude of the acceleration of the particle.
b) the distance from A to B.
c) the speed of the particle when it passes through B.

Question 4. A particle moves with constant acceleration $(4\mathbf{i} + 7\mathbf{j})$ ms^{-2}, and initially has velocity $(\mathbf{i} + 2\mathbf{j})$ ms^{-1}. Find the velocity of the particle at $t = 3$ seconds.

Question 5. A particle moves with constant acceleration $(-\mathbf{i} + 3\mathbf{j})$ ms^{-2}. It has velocity $(-2\mathbf{i} + 5\mathbf{j})$ ms^{-1} at $t = 7$ seconds. Find

a) the initial velocity of the particle.

b) the initial speed of the particle.

Question 6. An object is moving with constant acceleration. Initially it has velocity $\binom{4}{3}$ ms^{-1} and 7 seconds later it has velocity $\binom{8}{18}$ ms^{-1}.
Find the magnitude of its acceleration.

Question 7. Find the velocity vector of a particle in each case, given the speed v and direction in which the particle is travelling, leaving your answers in exact form:

a) $v = 10$ ms^{-1} parallel to $\begin{pmatrix}1\\0\end{pmatrix}$

b) $v = 4$ ms^{-1} parallel to $\begin{pmatrix}1\\1\end{pmatrix}$

c) $v = 0.3$ ms^{-1} parallel to $\begin{pmatrix}2\\-3\end{pmatrix}$

Question 8. The position vector, $\mathbf{r}$ metres, of an object at time t seconds is given by

$$\mathbf{r} = t^2\mathbf{i} + (3 - 2t)\mathbf{j}$$

Find its displacement between $t = 2$ and $t = 5$

Question 9. An object is initially at the origin moving with velocity $(2\mathbf{i} + 5\mathbf{j})$ ms^{-1}. It has a constant acceleration of $(6\mathbf{i} - \mathbf{j})$ ms^{-2}. Find

a) the position vector of the object at time t

b) the distance it is from the origin after 10 seconds.

Question 10. A particle P moves with constant acceleration $(5\mathbf{i} + 4\mathbf{j})$ ms^{-2}, having been initially at rest at initial position $(2\mathbf{i} - 7\mathbf{j})$ metres.

a) Show that the expression for the position vector of P at time t seconds is given by

$$\mathbf{r} = \left(2 + \frac{5}{2}t^2\right)\mathbf{i} + (2t^2 - 7)\mathbf{j}$$

b) Hence, find the distance the particle is from the origin after 2 seconds, giving your answer to 3 significant figures.

Question 11. Initially a particle position is $(6\mathbf{i} + 2\mathbf{j})$ m relative to a fixed origin O and is moving with velocity $(\mathbf{i} - 3\mathbf{j})$ ms^{-1} and constant acceleration $(8\mathbf{i} - 5\mathbf{j})$ ms^{-2}.
Find the position vector of the particle at $t = 4$ seconds.

Question 12. A particle moves with constant acceleration $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ ms^{-2} and passes through the point A with velocity $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$ ms^{-1}.
It then reaches point B 4 seconds after leaving A, where point B has position vector $\begin{pmatrix} 8 \\ -2 \end{pmatrix}$ metres relative to the origin.
Find the position vector of A.

Question 13. A particle moves with constant acceleration $(2\mathbf{i} - 4\mathbf{j})$ ms^{-2}. At $t = 0$ it passes through the origin and T seconds later it passes through a point with position vector $(-12\mathbf{i} + k\mathbf{j})$ metres at velocity $(4\mathbf{i} + 3\mathbf{j})$ ms^{-1}. Find

a) the value of T

b) the value of k

Question 14. A particle P moves in a straight line from position $\binom{5}{-2}$ metres to $\binom{k}{78}$ metres, moving parallel to the vector $\binom{7}{10}$. Find the value of k

Question 15. For each of the following, the two vectors represent the velocity of two particles which are moving with constant velocity and parallel to each other. In each case, find the value of k for this to hold:

a) $(2\mathbf{i} - 4\mathbf{j})$ ms^{-1} and $(4\mathbf{i} - k\mathbf{j})$ ms^{-1}

b) $(5\mathbf{i} + 3\mathbf{j})$ ms^{-1} and $(-10\mathbf{i} + k\mathbf{j})$ ms^{-1}

c) $(-3\mathbf{i} + 10\mathbf{j})$ ms^{-1} and $(k\mathbf{i} + 30\mathbf{j})$ ms^{-1}

d) $(1.5\mathbf{i} - 9\mathbf{j})$ ms^{-1} and $k(2\mathbf{i} - 12\mathbf{j})$ ms^{-1}

Question 16. A particle passes through the origin at $t = 0$ with velocity $\begin{pmatrix}9\\2\end{pmatrix}$ ms^{-1} and constant acceleration $\begin{pmatrix}2\\3\end{pmatrix}$ ms^{-2}. At $t = T$ the particle is moving in the direction of $\begin{pmatrix}4\\1\end{pmatrix}$

Find the value of T

Question 17. Emily is cycling with constant velocity $(4.2\mathbf{i} + 14.4\mathbf{j})$ ms^{-1} along a straight road. Henry is cycling parallel to Emily, on the same path.
Find the velocity for Henry in each of the following scenarios:

a) Henry is cycling twice as fast as Emily in the same direction.
b) Henry is cycling with speed 5 ms^{-1} in the same direction.
c) Henry is cycling with speed 12 ms^{-1} in the same direction.
d) Henry is cycling with speed 7.5 ms^{-1} in the opposite direction.

Question 18. Two particles A and B are moving parallel to each other. Particle A is moving with constant velocity $\begin{pmatrix} 9 \\ 21.6 \end{pmatrix}$ ms^{-1}. Particle B begins at a position of $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ m relative to a fixed origin and moves with constant speed 19.5 ms^{-1}.
Find

a) the speed of Particle A.
b) the velocity of Particle B.
c) the position vector of B after 5 seconds.

Question 19. Two particles A and B are moving parallel to each other, starting 4 metres apart and then at a fixed time later, are 4 metres apart again. Particle A travels a distance of 16 metres and particle B travels a distance of 20 metres.
Find the shortest distance between their two lines of motion, giving your answer to 3 significant figures.

Question 20. Two squirrels are running along parallel branches of trees in the woods. At time $t = 0$ seconds, the first squirrel starts at position $(-\mathbf{i} + 3\mathbf{j})$ metres and is moving with constant speed of 5.2 ms^{-1}. The second squirrel runs with constant velocity $(2.4\mathbf{i} + 5.76\mathbf{j})$ ms^{-1}.
By modelling the branches as flat straight paths, find

a) the speed of the second squirrel.

b) the position vector of the first squirrel when 2.5 seconds have passed.

Given that the second squirrel starts at position $(11\mathbf{i} - 2\mathbf{j})$ metres,

c) determine the shortest distance between their two branches.

Question 21. Particle A starts with position vector $\begin{pmatrix}-7\\-1\end{pmatrix}$ m and constant velocity $\begin{pmatrix}6\\3\end{pmatrix}$ ms^{-1}. Particle B starts with position vector $\begin{pmatrix}0\\2\end{pmatrix}$ m and constant velocity $\begin{pmatrix}2.5\\1.5\end{pmatrix}$ ms^{-1}.
Show that the particles collide.

Question 22. Particles A and B are both moving in straight lines. Particle A has a constant velocity of $\begin{pmatrix}3.5\\12\end{pmatrix}$ ms^{-1} and Particle B moves with a speed of 6.25 ms^{-1}.
Both particle A and particle B pass through point P. Particle B reaches point P first, with particle A passing through it 2 seconds later. Given that the two lines of motion are perpendicular to each other, find the distance between particle A and particle B 6 seconds after particle B passed through point P.

Question 23. The following questions that use i and j are unit vectors pointing east and north respectively. For each of the following velocity vectors, state the direction that the particle is moving in:

a) $(2\mathbf{i})$ ms^{-1}
b) $(-5\mathbf{j})$ ms^{-1}
c) $(4\mathbf{i} + 4\mathbf{j})$ ms^{-1}
d) $(-5\mathbf{i} + 5\mathbf{j})$ ms^{-1}

Question 24. A ship, moving with constant acceleration, is travelling due east at a speed of 4 ms^{-1}. After 40 seconds it is travelling due south at 2 ms^{-1}. Find

a) the displacement from its initial position after 40 seconds.
b) the distance the ship has travelled from its initial position after 40 seconds.

Question 25. The velocity, in ms^{-1}, of an object at time t seconds is given by

$$\mathbf{v} = (t-5)\mathbf{i} + (2t-13)\mathbf{j}$$

a) Show the object is never instantaneously at rest.
b) Find the time at which the object is moving directly to the south.
c) Find the time at which the object is moving to the north-east.

Question 26. An object moves so that its position vector, in metres, relative to the origin O at time t seconds is given by

$$\mathbf{r} = (105 + 8t)\mathbf{i} + t^2\mathbf{j}$$

a) Determine the time at which the object is east of the origin.
b) Determine the time at which the object is north-east of the origin.
c) Explain why the object never moves in the westwards direction.

Question 27. A particle passes point A with velocity $\binom{3}{8}$ ms^{-1} and has acceleration $\binom{2}{-20}$ ms^{-2}.
At the instant it reaches point B it is travelling in a due east direction.
Find the time at which it reaches point B

Question 28. A particle passes point A with velocity $\binom{1}{3}$ ms^{-1} and has acceleration $\binom{15}{5}$ ms^{-2}.
At the instant it reaches point B it is travelling in a due north-east direction.
Find the time at which it reaches point B.

Question 29. A particle is travelling with constant acceleration. It passes point A with velocity $\begin{pmatrix} 6 \\ -12 \end{pmatrix}$ ms^{-1} and then passes point B with velocity $\begin{pmatrix} -2 \\ 2 \end{pmatrix}$ ms^{-1}
Find

a) the acceleration of the particle when $t = 2$
b) the time at which it is travelling due south.
c) the speed at which it is travelling at the exact point it is travelling due south.

Question 30. A particle is travelling with constant acceleration $(-\mathbf{i} - 3\mathbf{j})$ ms^{-2} and passes the origin with velocity $(13\mathbf{i} - 5\mathbf{j})$ ms^{-1}. After a further 5 seconds have passed, the particle reaches point P. Find

a) the speed the particle is travelling in when it passes through point P giving your answer to 3 significant figures.
b) Find the bearing at which the particle is travelling when it passes through point P

Question 31. A helicopter is initially hovering above a field. It then sets off so that its acceleration is $(0.4\mathbf{i} + 0.2\mathbf{j})$ ms^{-2}.
The helicopter does not change its height as it moves.
Find

a) the speed of the helicopter 15 seconds after it leaves its position above the field.
b) the bearing on which the helicopter is travelling, 15 seconds after it leaves its position above the field, giving your answer correct to the nearest degree.

The helicopter stops accelerating when it is 240 metres from its initial position.

c) Find the time that it takes for the helicopter to travel from its initial position to the point where it stops accelerating.